

HARSH SHARMA

Data Engineer

Boston, MA | +1 (857) 294-3684 | Email: harshinfall24@gmail.com | [LinkedIn](#)

Professional Summary

Data Engineer with 5+ years of experience in building scalable data pipelines, optimizing big data architectures, and managing cloud-based solutions. Proficient in Python, SQL, Apache Spark, AWS, and real-time streaming technologies like Kafka and Kinesis. Strong expertise in data warehousing, ETL processes, and data modeling, with a focus on delivering high-quality insights and supporting machine learning workflows. Experienced in designing, building, and optimizing data pipelines and architectures for large-scale data processing to deliver high-performance solutions that drive business insights.

Professional Experience

Bankers Health Group, NY, USA

Feb 2024 – Present

Data Engineer

- Optimized data pipelines in Python, SQL, and PySpark with Apache Spark, handling over 1TB of daily data and improving overall processing speed by 30%.
- Automated ETL processes using Apache Airflow and Talend, reducing manual intervention by 40% and improving data flow reliability.
- Maintained data architectures on AWS (S3, EC2, Glue, Lambda, Redshift), achieving a 25% cost reduction through optimized resource allocation and storage.
- Improved MySQL, PostgreSQL, and HBase databases, reducing query execution times by 35% and enhancing data retrieval efficiency for analytics.
- Facilitated data warehousing efforts using Snowflake and Google BigQuery, leading to a 20% improvement in data accessibility and reporting speed for business stakeholders.
- Integrated real-time data streaming with Apache Kafka and AWS Kinesis, decreasing data processing latency by 20% for time-sensitive applications.
- Implemented best practices for data modeling, normalization, and warehousing, ensuring clean, structured, and high-quality datasets, improving data integrity by 10%.
- Adhered to Agile and Kanban methodologies, delivering features on time with a 90% sprint success rate.

Clear Sky IT Solution, NY, USA

Jan 2023 – June 2023

Data Engineer (Internship)

- Designed and Developed scalable data pipelines to process large volumes of structured and unstructured data using tools like Python, SQL, and Spark.
- Coordinated in building and optimizing data warehouse solutions to support analytics and business intelligence teams.
- Developed ETL pipelines using Python, SQL, and Apache Spark, processing large datasets from extraordinary sources into centralized databases.
- Guided in the integration of raw data from sources into centralized databases, ensuring data accuracy and consistency.
- Conducted data cleaning, transformation, and validation using Pandas, NumPy, and Airflow, ensuring data integrity and quality for analytics.
- Collaborated with cross-functional teams to gather and understand data requirements and implement appropriate solutions.
- Construct interactive dashboards and visualizations using Tableau, Power BI, and Looker, providing actionable insights to stakeholders.
- Created technical documentation for data workflows, pipelines, and system configurations.
- Transformed cloud platforms such as AWS or Azure, to deploy and manage data solutions in a secure and scalable environment.
- Gained hands-on experience with tools and frameworks such as Hadoop, Kafka, Airflow, and Tableau to support real-time and batch data processing needs.

Yatra Online PVT LTD, India

Nov 2017 – August 2021

Data Engineer

- Engineered robust data pipelines utilizing Apache Spark and PySpark, achieving a 30% increase in processing speed: enhanced workflow efficiency across the data engineering team by reducing manual intervention.
- Enhanced data retrieval and storage by designing efficient SQL queries and indexing strategies in MySQL and PostgreSQL, resulting in a 30% boost in query performance.
- Partnered with data scientists to preprocess and structure data for machine learning models using Scikit-Learn, TensorFlow, and Keras, resulting in a 15% increase in model accuracy.
- Employed Docker and Kubernetes for containerizing and orchestrating data applications, reducing deployment times by 20% and improving scalability.
- Created real-time data dashboards using Power BI and Tableau, increasing insight delivery speed by 25% and improving reporting

accuracy for stakeholders.

- Influenced in implementing Azure Synapse and Databricks solutions, enhancing data analytics capabilities and improving query performance by 20%.

Skills

- **Programming Languages:** Python, SQL, PySpark, C, Scala
- **Big Data Technologies:** Apache Spark, Hadoop, Kafta, AWS Kinesis, AWS EC2, AWS S3
- **BI Tools:** Tableau, Power BI
- **Data Warehousing:** Amazon Redshift, Google Big Query, Azure Data Factory, Azure Databricks, Azure Synapse, Snowflake
- **Database Management:** MySQL, PostgreSQL, HBase, Cosmos DB, Snowflake, MongoDB
- **ETL Tools:** Apache Airflow, Talend, Informatica
- **Cloud Platforms:** AWS, AWS GLUE, AWS Redshift, AWS Lambda, Google Cloud
- **Streaming Analytics:** Apache Kafka, Apache Flink
- **Containerization and Orchestration:** Docker, Kubernetes, Apache
- **Methodologies:** Agile, Scrum, Waterfall, Kanban
- **Data Skills:** Visualization, Data Modeling, Data Normalization, Data Warehousing, Data Mining, Data Analysis, Statistic
- **Machine Learning:** Scikit-Learn, TensorFlow, Keras
- **Frameworks:** Pandas, NumPy, Dask

Education

Master of Science in Information Systems

December - 2023

North-eastern University, MA

Bachelor of Engineering in Computer Science

May - 2018

AMITY University, India

Certification: Applied Data Science with Python – Level 2, IBM

August - 2019

Projects:

1. CIFAR – 10 Data Image Classification Using CNN

- Built a CNN model for vehicle detection and tracking, achieving a validation accuracy of 62.4% by applying data transformation and model optimization techniques.

2. Healthcare Drug Distribution System

- Delineate end-to-end data models and schemas using SQL for a unified drug distribution system, ensuring operational consistency.
- Implemented Oracle 21c function for efficient data handling and data warehouse management.